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Pancracine as a potential agent in the treatment of oncological diseases



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INTRODUCTION

Nearly 600 Amaryllidaceae alkaloids (AmA) have been detected in plants of many different species belonging to the Amaryllidaceae family. These alkaloids show wide range of biological activities. They are isolated from plant material (mainly from bulbs) and tested for possible use in treatment of various illnesses. Among the most important biological activities of AmA belong those associated with Alzheimer's disease and the cytotoxic activity. These and other diseases are becoming increasingly widespread all over the world. From this reason the development of new potential drugs is needed. Review from Newman et al. (2016)¹ shows that 51 % of all drugs registered between 1981 and 2014 and 65 % of small-molecule drugs are related to natural products (unaltered natural products and their derivatives, synthetic drugs with NP pharmacophore, etc.).

The most studied AmA regarding to cytotoxic/antineoplastic activity are narciclassine (inhibitor of mitosis and proteosynthesis), pancratistatine, haemanthamine and lycorine (inductors of apoptosis, inhibitors of cell cycle). These alkaloids also serve as templates for a synthesis of their semisynthetic analogues. Among the most widely used templates of AmA belong lycorine and haemanthamine, because pancratistatine and narciclassine do not occur in plant material in sufficient amount.

Narcissus cv. PROFESSOR EINSTEIN was chosen from extensive screening for further phytochemical work because it contains substances which haven't been isolated and tested at our department yet. Twenty-five substances were isolated in pure form, and identified by MS, HRMS and 1D- and 2D-NMR techniques and X-ray as caranine, 9-O-demethylmaritidine, epimaritidine, epinorgalanthamine, eugenine, galanthamine, galanthine, incartine, ismine, haemanthamine, hippeastrine, homolycorine, 9-O-demethylhomolycorine, lycorine, lycoramine, lycoraminone, masonine, 9-O-methylpseudolycorine isomer, narwedine, norlycoramine, norpluviine, 7-oxonorpluviine, pancracine, pluviine and tazettine. All alkaloids were depending on obtained amount tested on the inhibitory activity against different enzymes associated with Alzheimer's disease (HuAChE, HuBuChE, POP, GSK-3 β), AKR1C3 (enzyme associated with prostate cancer in men), **various cancer cell lines** (Jurkat, MOLT-4, A549, HT-29, HeLa, MCF-7, Caco-2, etc.) and diverse infectious agents - bacteria, fungi, viruses, protozoa (*Plasmodium* spp. liverstage).

EXPERIMENTAL PART

Isolation of alkaloids

- 34 kg of fresh bulbs boiled 2 $\times$ in 96% EtOH (20 min)
- liquid-liquid extraction from summary ethanolic extract
- column chromatography (Al₂O₃ deactivated with 6 % H₂O; light petrol-CHCl₃-EtOH)
- analytical and preparative TLC
- crystallization

Identification of alkaloids:

- NIST 11, Wiley, 1D- and 2D-NMR techniques, X-ray, standards, literature

Cytotoxicity assay:

reader Tecan Infinite 2000, GraphPad Prism 5
3 independent experiments

WST-1 assay single-dose screening

- cells were incubated in 96-well plate for 24 hours (seeding density 1.10³ to 50.10³ cells per well)
- cells were incubated for 48 hours with tested compounds (c = 10 μ mol/L)
- standard **doxorubicine** (c = 1 μ mol/L)
- growth percentage compared to control (MRC-5, healthy cells)

WST-1 assay IC₅₀ determination

- tested concentrations 0.1 - 100 μ M

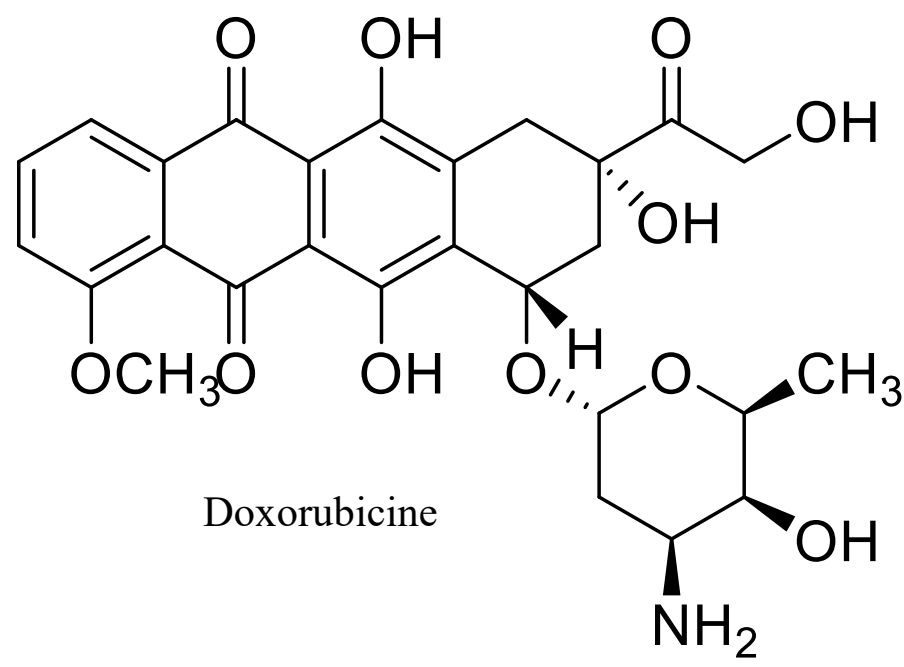
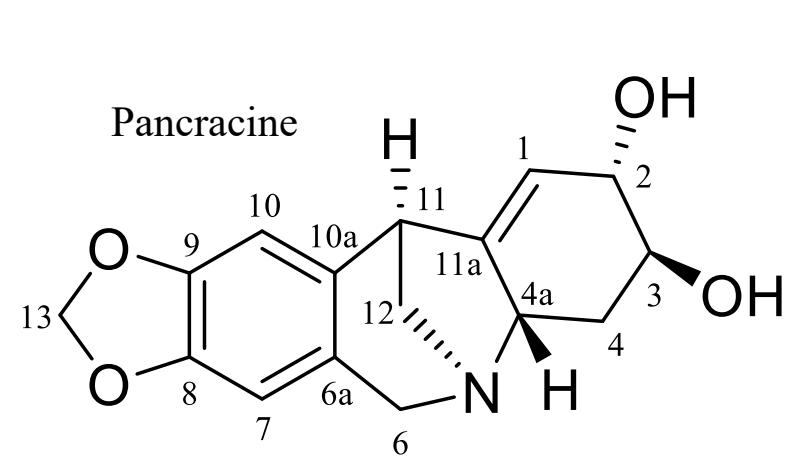
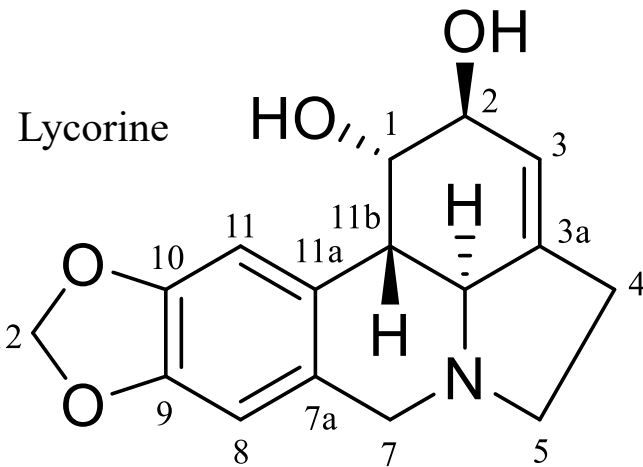
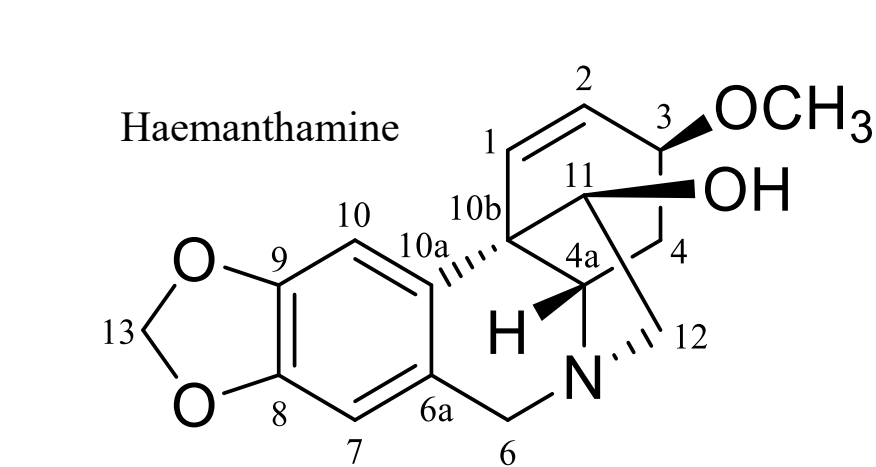
SCREENING OF CYTOTOXIC ACTIVITY

Compound ¹	Jurkat	MOLT-4	A549	HT-29	PANC1	A2780	HeLa	MCF-7	SAOS-2	MRC-5 ¹
Masonine	76	105	109	110	94	102	117	98	98	99
Homolycorine	78	110	114	199	100	96	108	102	97	97
Ismine	-	-	-	-	-	-	-	-	-	-
Caranine	99	99	111	58	82	102	68	99	77	-
Galanthamine	94	89	94	74	81	98	82	94	100	-
Narwedine	115	117	96	107	107	90	105	113	100	-
Lycoraminone	102	99	107	98	97	100	98	95	102	96
Pluviine	-	-	-	-	-	-	-	-	-	-
Norlycoramine	102	106	93	94	100	109	93	99	102	108
Incartine	-	-	-	-	-	-	-	-	-	-
Galanthine	98	103	93	64	97	163	93	114	97	-
Lycoramine	102	105	96	83	102	92	82	101	97	-
Epinorgalanthamine	-	-	-	-	-	-	-	-	-	-
Haemanthamine	18	6	33	5	37	38	16	17	34	32
Hippeastrine	40	50	69	66	84	50	86	70	83	69
Epimaritidine	102	106	110	103	91	98	97	96	103	98
9-O-Methylpseudolycorine isomer	97	99	87	88	90	95	98	88	93	96
Lycorine	5	9	27	5	36	39	15	27	27	25
Tazettine	93	95	83	57	76	101	84	94	92	-
Eugenine	-	-	-	-	-	-	-	-	-	-
Norpluviine	96	107	111	92	84	101	91	100	96	92
9-O-Demethylmaritidine	97	95	104	89	80	98	93	91	95	92
Pancracine	17	1	29	39	52	40	28	18	26	39
9-O-Demethylhomolycorine	100	100	109	106	92	94	96	102	103	93
7-Oxonorpluviine	-	-	-	-	-	-	-	-	-	-
Doxorubicine ²	0	0	66	77	59	5	7	41	73	40
DMSO (0.1%) ¹	113	115	106	87	95	98	89	99	102	-

¹c = 10 μ mol/L; ²standard, c = 1 μ mol/L; ³control – healthy cells; ⁴negative control - DMSO

Cell viability compared to control after 48 hours of incubation [%]

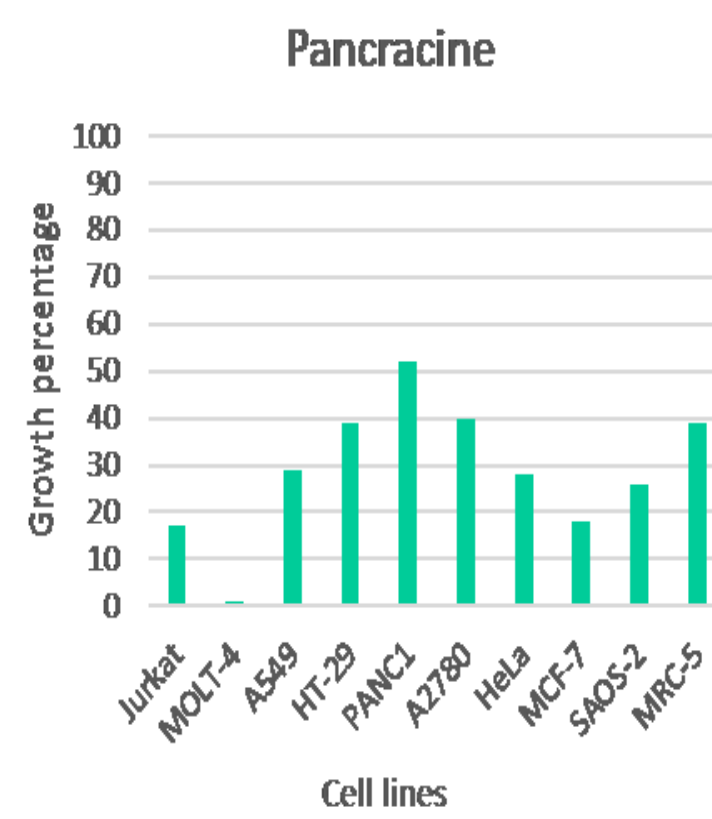
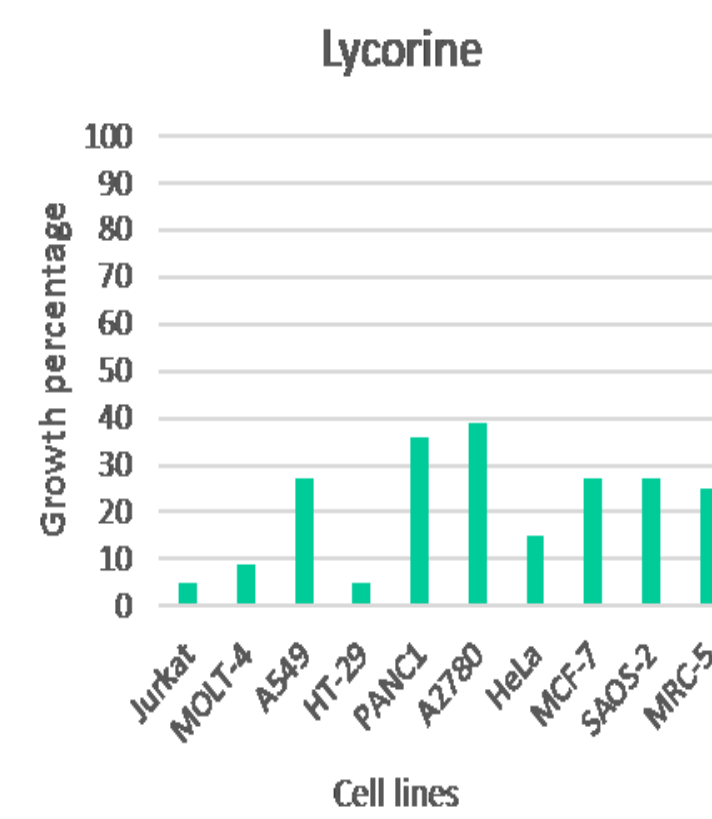
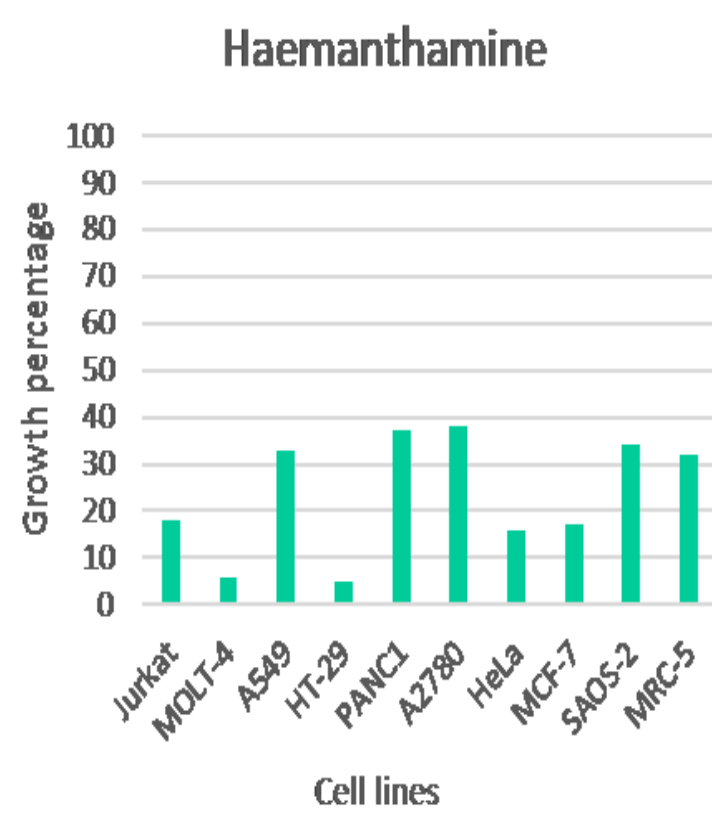
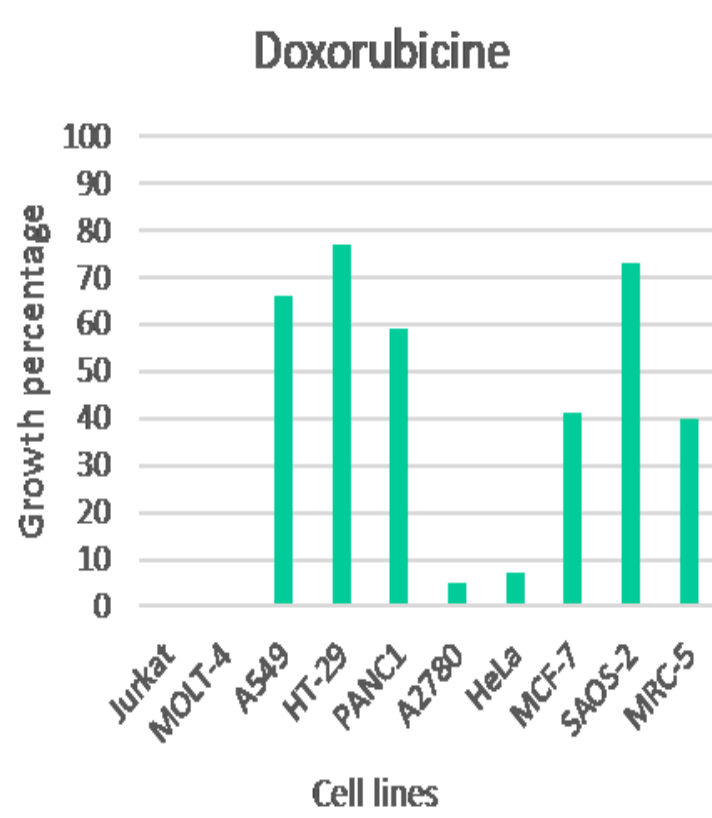
0 - 25	26 - 50	51 - 75	76 - 100
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CONCLUSION

From twenty-five isolated alkaloids were 19 isolated in sufficient amount to be screened on 10 cell lines panel for their cytotoxic activity. Hippeastrine showed mild activity against some cell lines but its IC₅₀ was not tested. Haemanthamine, lycorine and pancracine have shown significant activity. Haemanthamine and lycorine are very well known and they serve as lead structures for preparation of various semisynthetic derivatives. Pancracine has not been intensively studied so far. Its cytotoxic activity seems good so its IC₅₀ against 9 from 10 cell lines panel was tested. The best activity was performed against the SAOS-2, A549 and HT-29 cell lines. Currently, pancracine has been tested to clarify its mechanism of action, induction of apoptosis and interaction with cell cycle. Pancracine will also serve as a template structure for the semi-synthesis of its derivatives.

The most active screened alkaloids



Doxorubicine = standard, c = 1 μ mol/L; alkaloids c = 10 μ mol/L

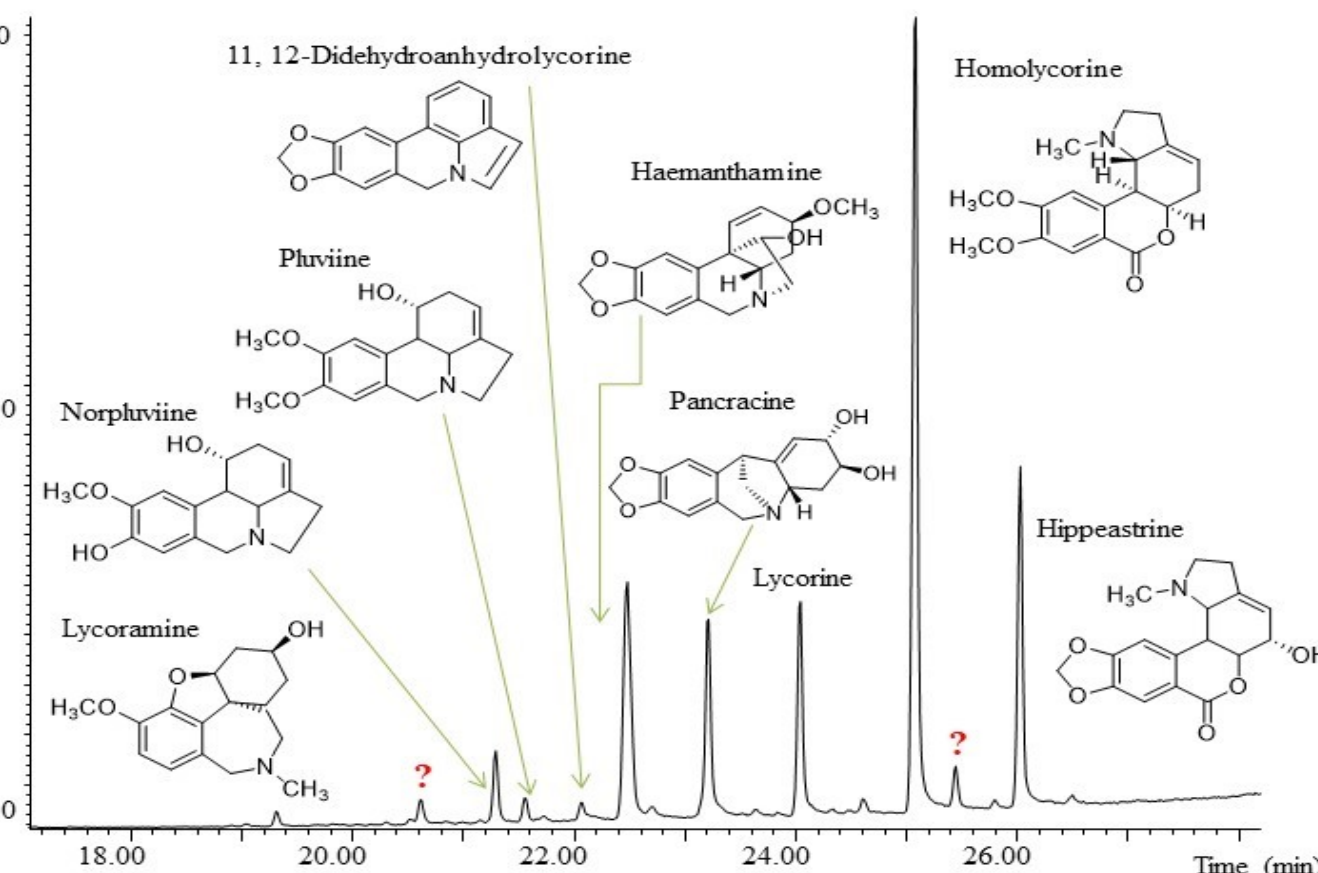
the most active compounds



IC₅₀ [μ M]



GC-MS analysis of NPE summary extract



SCREENED CELL LINES

- Jurkat - acute T cell leukemia
- MOLT-4 - acute lymphoblastic leukemia
- A549 - lung adenocarcinoma
- HT-29 - colorectal adenocarcinoma
- PANC-1 - pancreas epithelioid carcinoma
- A2780 - ovarian carcinoma
- HeLa - cervix adenocarcinoma
- MCF-7 - breast adenocarcinoma
- SAOS-2 - osteosarcoma
- MRC-5 - normal lung fibroblasts
- NHDF - normal human dermal fibroblasts

Compound	Mean GP ¹	Range of GP	Most sensitive cell lines	% of viable cells
Haemanthamine	27	5 - 38	HT-29, MOLT-4, HeLa	5, 6, 16
Lycorine	22	5 - 39	Jurkat, HT-29, MOLT-4	5, 5, 9
Pancracine	29	1 - 52	MOLT-4, Jurkat, MCF-7	1, 17, 18
Doxorubicine	24	0 - 77	MOLT-4, Jurkat, A2780	0, 0, 5

¹ Mean growth percent (GP) value was calculated for each compound as an average of 10 cell lines proliferation in percent.

Cell line	Haemanthamine	Lycorine	Pancracine	Doxorubicine ¹
Jurkat	1.4 $\pm$ 0.3	1.3 $\pm$ 0.1	5.1 $\pm$ 0.3	0.05 $\pm$ 0.02
MOLT-4	1.2 $\pm$ 0.1	1.0 $\pm$ 0.2	2.7 $\pm$ 0.3	<0.01
A549	1.1 $\pm$ 0.2	1.0 $\pm$ 0.1	2.3 $\pm$ 0.4	0.43 $\pm$ 0.06
HT-29	0.3 $\pm$ 0.1	0.8 $\pm$ 0.3	2.6 $\pm$ 0.5	0.77 $\pm$ 0.24
PANC1	9.8 $\pm$ 0.4	0.7 $\pm$ 0.4	n.t.	n.t.
A2780	0.7 $\pm$ 0.4	1.2 $\pm$ 0.1	5.1 $\pm$ 0.4	<0.01
HeLa	0.6 $\pm$ 0.1	0.9 $\pm$ 0.3	5.0 $\pm$ 0.3	0.55 $\pm$ 0.05
MCF-7	0.8 $\pm$ 0.1	0.8 $\pm$ 0.5	2.7 $\pm$ 0.4	0.44 $\pm$ 0.10
SAOS-2	1.1 $\pm$ 0.4	1.2 $\pm$ 0.4	2.2 $\pm$ 0.3	0.10 $\pm$ 0.17
MRC-5 ²	n.t.	n.t.	5.2 $\pm$ 0.3	0.72 $\pm$ 0.23
NHDF ²	0.5 $\pm$ 0.1	0.8 $\pm$ 0.3	n.t.	n.t.

¹standard; ²control – healthy cells

LITERATURE

¹ Newman D. J., Cragg G. M. (2016) Natural Products as Sources of New Drugs from 1981 to 2014. Journal of Natural Products. 79, 629-661.

ACKNOWLEDGEMENT

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